Applied Linear Algebra in Data Analysis

Eigenvalues and Eigenvectors Applications

Sivakumar Balasubramanian

Christian Medical College Vellore
Department of Bioengineering

Recap: Eigenvalues and Eigenvectors

Matrices $\mathbf{A} \in \mathbb{R}^{p \times q}$ represent linear transformations $T : \mathbb{R}^q \to \mathbb{R}^p$, and they are basis dependent.

Any linear transformation represented by $\mathbf{A} \in \mathbb{C}^{n \times n}$ has vectors that are only scaled by the transformation,

$$\mathbf{A}\mathbf{x} = \lambda \mathbf{x}, \quad \mathbf{x} \in \mathbb{C}^n, \ \lambda \in \mathbb{C}, \ \ \mathbf{x} \neq \mathbf{0}$$

$(\lambda, \mathbf{x})$ is called the eigenpair of $\mathbf{A}$.

A matrix $\mathbf{A}$ with a full set of linearly independent eigenvectors can be "diagonalized" to a diagonal matrix Λ,

$$\mathbf{A} = \mathbf{X}\Lambda\mathbf{X}^{-1}$$

When there isn't a full set of linearly independent eigenvectors, it can only be block diagonalized,

$$\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$$

Dynamical Systems

Dynamical systems evolve over time, which generally have memory associated with them.

The output of a system at any time instant will depend on the current input and the current "state" of the system.

State is any information that is necessary to determine the future beahvior of the system given the future inputs.

The state space representation of a general dynamical system is given by,

Continuous-time	Discrete-time
$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t))$	$\mathbf{x}[n+1] = f(\mathbf{x}[n], \mathbf{u}[n])$
$\mathbf{y}(t) = g(\mathbf{x}(t), \mathbf{u}(t))$	$\mathbf{y}[n] = g(\mathbf{x}[n], \mathbf{u}[n])$

Linear Dynamical Systems

Continuous-time linear dynamical system:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \longrightarrow \text{State equation}$$
$$\dot{\mathbf{y}}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \longrightarrow \text{Measurement equation}$$

where,

- $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the *system* matrix.
- $\mathbf{B} \in \mathbb{R}^{n \times p}$ is the *input* matrix.
- $\mathbf{C} \in \mathbb{R}^{m \times n}$ is the *output* matrix.
- $\mathbf{D} \in \mathbb{R}^{m \times p}$ is the *feedforward* matrix.

Discrete-time linear dynamical system:

$$\mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n] + \mathbf{B}\mathbf{u}[n]$$
$$\mathbf{y}[n] = \mathbf{C}\mathbf{x}[n] + \mathbf{D}\mathbf{u}[n]$$

Discrete-time linear dynamical system

Zero-input response of a discrete-time linear dynamical system is given by,

$$\mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n] \implies \mathbf{x}[n] = \mathbf{A}^n\mathbf{x}[0]$$

where, $\mathbf{x}[0]$ is the initial state of the system.

When there is a non-zero input, the response is given by,

$$\mathbf{x}[n+1] = \mathbf{A}\mathbf{x}[n] + \mathbf{B}\mathbf{u}[n] \implies \mathbf{x}[n] = \mathbf{A}^n\mathbf{x}[0] + \sum_{k=0}^{n-1} \mathbf{A}^{n-k-1}\mathbf{B}\mathbf{u}[k]$$

Discrete-time linear dynamical system: Eigen Decomposition

Let $\mathbf{A} = \mathbf{X}\Lambda\mathbf{X}^{-1}$, then we have,

$$\mathbf{x}\left[n+1\right] = \mathbf{A}\mathbf{x}\left[n\right] \implies \mathbf{x}_{\mathcal{V}}\left[n+1\right] = \Lambda\mathbf{x}_{\mathcal{V}}\left[n\right]$$

where, $\mathbf{x}_{\mathcal{V}} = \mathbf{X}^{-1}\mathbf{x}$ is the representation of $\mathbf{x}$ in the eigenbasis.

In the eigenbasis, the system is decoupled into n independent scalar systems,

The components of the state in the eigenbasis evolve independently of each other, and the evolution is given by,

$$x_{\mathcal{V},i}\left[n+1\right] = \lambda_i x_{\mathcal{V},i}\left[n\right] \implies x_{\mathcal{V},i}\left[n\right] = \lambda_i^n x_{\mathcal{V},i}\left[0\right]$$

Continuous-time linear dynamical system

Zero-input response of a continuous-time linear dynamical system is given by,

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) \implies \mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$$

where, $\mathbf{x}(0)$ is the initial state of the system.

When there is a non-zero input, the response is given by,

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \implies \mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau$$

Continuous-time linear dynamical system: Eigen Decomposition

Let $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$, then we have,

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) \implies \mathbf{x}_{\mathcal{V}}(t) = \mathbf{\Lambda}\mathbf{x}_{\mathcal{V}}(t)$$

where, $\mathbf{x}_{\mathcal{V}} = \mathbf{V}^{-1}\mathbf{x}$ is the representation of $\mathbf{x}$ in the eigenbasis.

In the eigenbasis, the system is decoupled into n independent scalar systems,

The components of the state in the eigenbasis evolve independently of each other, and the evolution is given by,

$$\dot{x_{\mathcal{V},i}}(t) = \lambda_i x_{\mathcal{V},i}(t) \implies x_{\mathcal{V},i}(t) = e^{\lambda_i t} x_{\mathcal{V},i}(0)$$

Linear dynamical system: Stability

Internal stability: Deals with the stability of the zero-input response of the system states, i.e. $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t))$ or $\mathbf{x}[n+1] = \mathbf{f}(\mathbf{x}[n])$.

An *equilibrium point* $\mathbf{x}_e$ of this system is defined as a point in the state space where,

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}_e) = \mathbf{0} \qquad \mathbf{x}[n+1] = \mathbf{f}(\mathbf{x}_e) = \mathbf{x}_e$$

i.e. if the system starts in this state, it stays in that state for all time.

Non-linear systems can have multiple equilibrium points, and the stability of these equilibrium points can be different.

Linear systems have equilibrium points in subspaces of the state space, and the stability of these equilibrium points are the same.

Internal Stability

Definition of stability in the Lyapunov sense for linear systems:

The zero-input response of a linear system $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$ is **stable or marginally stable** if every finite initial condition $\mathbf{x}(0^-)$ results in a bounded state trajectory $\mathbf{x}(t)$ $\forall t \geq 0$.

$$\|\mathbf{x}(t)\| \leq d, \quad \forall t \geq 0$$

The zero-input response is **asymtotically stable** if everyf initial condition $\mathbf{x}(0^-)$ results in a bounded state trajectory $\mathbf{x}(t)$ that coverges to 0 as $t \longrightarrow \infty$.

$$\|\mathbf{x}(t)\| \leq d \text{ and } \lim_{t \longrightarrow \infty} \|\mathbf{x}(t)\| = 0$$

Internal Stability

Continuous-time linear dynamical system.
$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$ is **marginally stable** $\iff$ eigenvales of $\mathbf{A}$ have either zero or negative real parts, and the eigenvalues with zero real parts have the same algebraic and geometric multiplicity.

$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$ is **asymptotically stable** $\iff$ all eigenvales of $\mathbf{A}$ have negative real parts.

When $\mathbf{A}$ is diagonalizable,

$$\|\mathbf{x}(t)\| = \|e^{t\mathbf{A}}\mathbf{x}(0^-)\| \leq \|e^{t\Lambda}\|\|\mathbf{x}(0^-)\|$$

- $\|\mathbf{x}(t)\| \leq e^{\sigma t}\|\mathbf{x}(0^-)\|$, where $\sigma = \max_i \operatorname{Re}\{\lambda_i\}$.
- $\sigma = 0$, $\|\mathbf{x}(t)\|$ is bounded $\forall t \geq 0$.
- $\sigma < 0$, $\lim_{t\to\infty} \|\mathbf{x}(t)\| = 0$.

Internal Stability

Continuous-time linear dynamical system.
When $\mathbf{A}$ is not diagonalizable, $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$,

- Consider the ith Jordan block, $\mathbf{J}_i = \lambda_i \mathbf{I} + \mathbf{N}$, Thus, $e^{t\mathbf{J}_i} = e^{\lambda_i t \mathbf{I}} e^{t\mathbf{N}}$

$$\|\mathbf{x}(t)\| \leq e^{\sigma_i t} \|e^{t\mathbf{N}}\| \|\mathbf{x}(0^-)\|$$

  where, $\sigma_i = \mathrm{Re}\{\lambda_i\}$, and $\mathbf{N}$ is a nilpotent matrix.
- When $\sigma_i = 0$, $\|e^{t\mathbf{N}}\|$ grows with time, and thus $\mathbf{x}(t)$ is not bounded.
- When $\sigma_i < 0$, the $e^{\sigma_i t}$ term does not allow $\mathbf{x}(t)$ to grow.

Internal Stability

Discrete-time linear dynamical system.
$\mathbf{x}[k+1] = \mathbf{A}\mathbf{x}[k]$ is **marginally stable** $\iff$ eigenvales of $\mathbf{A}$ have magnitude less than or equal to 1, and the eigenvalues with zero real parts have the same algebraic and geometric multiplicity.

$\mathbf{x}[k+1] = \mathbf{A}\mathbf{x}[k]$ is **asymptotically stable** $\iff$ all eigenvales of $\mathbf{A}$ have magnitude less than 1.

When $\mathbf{A}$ is diagonalizable (λ_i are the eigenvalues of $\mathbf{A}$),

- $\|\mathbf{x}[k]\| \leq |\lambda|^k \|\mathbf{x}[0]\|$, where $\lambda = \max_i |\lambda_i|$.
- $|\lambda| = 1$, $\|\mathbf{x}[k]\|$ is bounded $\forall k > 0$.
- $|\lambda| < 1$, $\lim_{k\to\infty} \|\mathbf{x}[k]\| = 0$.

Internal Stability

When $\mathbf{A}$ is not diagonalizable, $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$,

- Consider the ith Jordan block, $\mathbf{J}_i^k = (\lambda_i \mathbf{I} + \mathbf{N})^k = \sum_{l=0}^{k} \frac{k!}{(k-l)!l!} \lambda_i^l \mathbf{N}^{k-l}$
- When $|\lambda_i| = 1$, $\|\mathbf{J}_i^k\|$ grows with time, and thus $\mathbf{x}[k]$ is not bounded.
- When $|\lambda_i| < 1$, the λ_i^l term does not allow $\mathbf{x}[k]$ to grow.

Linear Approximation of Nonlinear Systems

Consider a simple pendulum, which is a nonlinear system. The state space representation of the pendulum is given by,

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{\theta}(t) \\ \ddot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} x_2(t) \\ -\frac{g}{l} \sin x_1(t) \end{bmatrix}$$

where, θ is the angle of the pendulum with respect to the vertical, g is the acceleration due to gravity, and l is the length of the pendulum.

Where are the equilibrium points of this system?

Linear Approximation of Nonlinear Systems

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{\theta}(t) \\ \ddot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} x_2(t) \\ -\frac{g}{l}\sin x_1(t) \end{bmatrix}$$

We can evaluate the stability of the equilibrium points by linearizing the system around these points. The Jacobian of the system is given by,

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}$$

where, $f_1 = x_2$ and $f_2 = -\frac{g}{l}\sin x_1$.

This gives us the linear approximation of the system around a point,

$$\dot{\mathbf{x}}(t) \approx \mathbf{J}(\mathbf{x})\mathbf{x}(t) = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l}\cos x_1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Eigenvalues of $\mathbf{J}(\mathbf{x})$ can tell us about the stability of an equilibrium point.